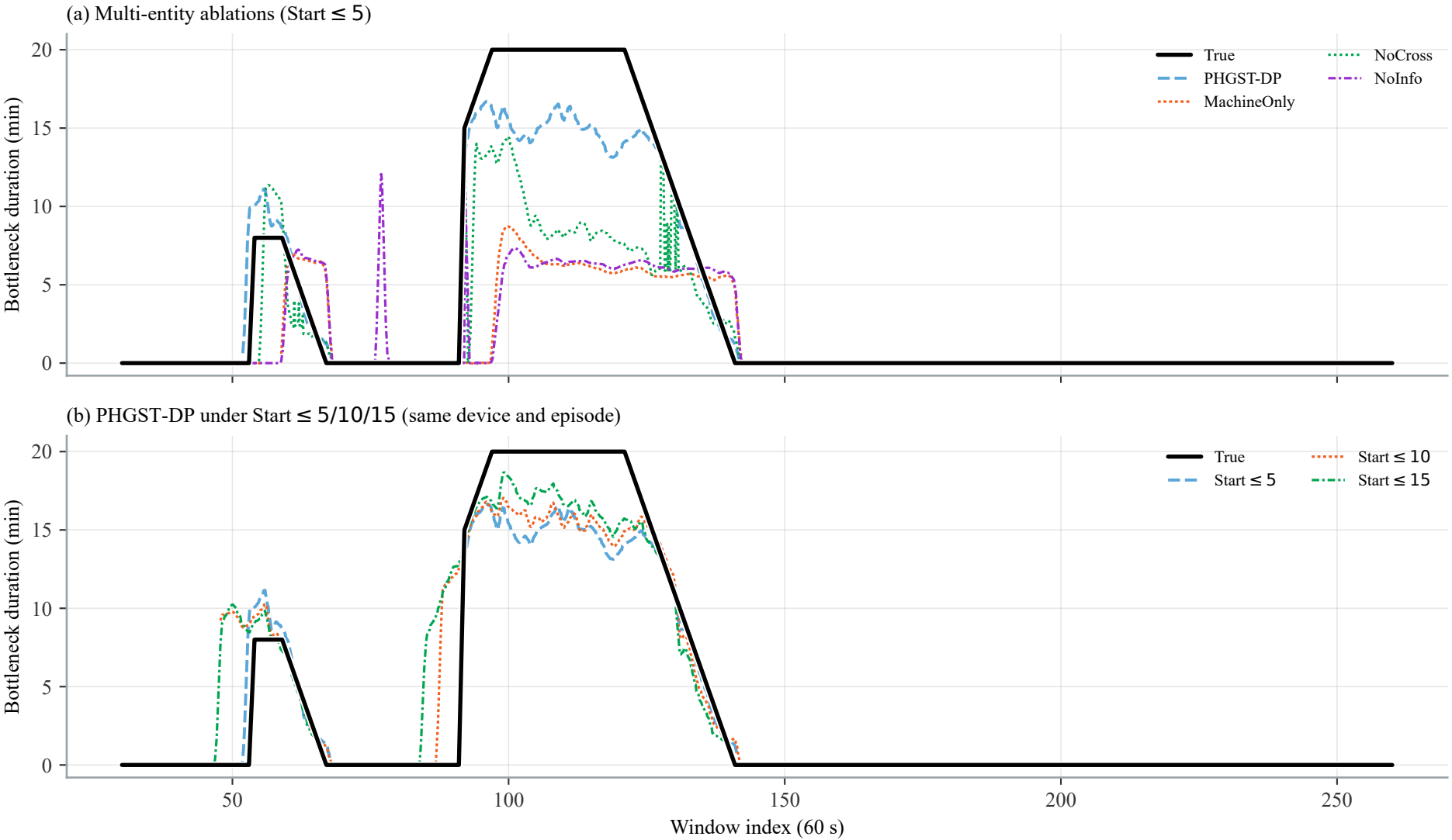

(a) Multi-entity ablations (Start ≤ 5)
Bottleneck duration (min)
0
5
10
15
20
True
PHGST-DP
MachineOnly
NoCross
NoInfo
(b) PHGST-DP under Start ≤ 5/10/15 (same device and episode)
Bottleneck duration (min)
0
5
10
15
20
True
Start ≤ 5
Start ≤ 10
Start ≤ 15
50
100
150
200
250
Window index (60 s)